

Adidas Sales Analytics

1. Project Overview

This project analyzes 9,648 Adidas sales transactions recorded from 01 Jan 2020 to 31 Dec 2021. The objective is to transform transactional sales data into an interactive business intelligence solution that explains revenue, profitability, retailer performance, product-category performance, sales channels, pricing behavior and future opportunities.

The solution follows an end-to-end analytics workflow: Excel data → Python/Pandas preprocessing → feature engineering → PostgreSQL storage → Power BI data model and DAX → interactive dashboard → forecasting and what-if scenario analysis.

2. Dataset Summary

Rows 9,648 transactions	Columns 14 original fields	Total Sales \$120.17M source total	Operating Profit \$47.22M source total	Units Sold 2.48M units
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Dimension	Count / Range	Purpose
Retailers	6	Compare channel partners and revenue contribution
Regions / States / Cities	5 / 50 / 52	Geographic performance analysis
Product Categories	3	Street Footwear, Apparel, Athletic Footwear
Sales Methods	3	Online, Outlet, In-store
Gender Types	2	Men and Women sales comparison
Invoice Period	01 Jan 2020 – 31 Dec 2021	Time-series and year-over-year analysis

3. Exploratory Data Analysis using Python

- Loaded the Excel dataset into Pandas and inspected the structure with head(), info() and describe().
- Validated data quality: 0 missing values across the 14 columns and 0 duplicate rows in the source workbook.
- Reviewed categorical dimensions, date coverage and numeric ranges before building derived features.

Key analytical baseline

Total sales are \$120.17M, operating profit is \$47.22M, and the weighted profit margin is 39.3%. Average sales per unit are approximately \$48.48, while profit per unit is approximately \$19.05.

Data Preprocessing & Feature Engineering

4. Data Preprocessing using Python

The preprocessing notebook was used to prepare the raw transaction data for consistent downstream analysis. The process focused on reproducibility, clean naming, correct date handling and analytical feature creation.

Step	What was done
Data Loading	Imported Adidas_Sales_Data.xlsx into a Pandas DataFrame.
Data Inspection	Used df.head(), df.info() and df.describe(include="all") to understand structure, data types and distributions.
Data Quality Validation	Checked every column for null values and duplicates. The source contained 0 nulls and 0 duplicate rows.
Column Standardization	Converted column names to lowercase and replaced spaces with underscores, creating consistent names such as invoice_date, total_sales and operating_profit.
Date Conversion	Converted invoice_date to Pandas datetime format so the data could support time-based analysis.
Date Feature Engineering	Created year, month, day, day_of_week, day_name, week_of_year, quarter and is_weekend.
Revenue Feature Engineering	Created profit_per_unit and sales_per_unit to support pricing and profitability analysis.

Why preprocessing matters

Preprocessing makes the downstream Power BI model more reliable. Standardized names reduce ambiguity, datetime conversion enables time intelligence, and derived unit-level metrics make pricing and margin questions easier to answer.

5. Database Integration

The cleaned DataFrame was loaded into a PostgreSQL database using SQLAlchemy and the PostgreSQL driver. The notebook writes the processed table to the database so that the analytical layer can work from a structured relational source rather than repeatedly reading the raw Excel workbook.

For security, database credentials used during local development are intentionally not included in this report.

6. Prepared Analytical Dataset

- Processed table: customer in the local Adidas_Sales database.
- Analytical grain: one sales transaction per row; 9,648 rows after preprocessing.
- Prepared dimensions: retailer, region, state, city, gender type, product category and sales method.
- Prepared measures: sales, operating profit, units sold, operating margin, sales per unit and profit per unit.

Business Analysis & Key Performance Findings

7. Business Questions Answered

Question	Analytical purpose
Revenue performance	How much sales and operating profit were generated overall, and how does performance change over time?
Retailer contribution	Which retail partners contribute the most revenue, and how concentrated is the business?
Product performance	Which product categories generate the highest sales, units and profit?
Geographic performance	Which regions and cities contribute the most sales?
Sales method	Which selling channel—Online, Outlet or In-store—drives the greatest revenue?
Pricing & profitability	How are price per unit, units sold, profit and margin related?
Future planning	What happens to estimated sales when price or sales volume assumptions are changed?

8. Key Performance Findings

- Overall: \$120.17M total sales, \$47.22M operating profit and 2.48M units sold, giving a weighted margin of 39.3%.
- Category leader: Street Footwear generated \$44.88M (37.4%) and \$18.12M operating profit.
- Retailer leader: West Gear generated \$32.41M, followed by Foot Locker at \$29.02M and Sports Direct at \$24.62M.
- Retailer concentration: the top three retailers together contribute 71.6% of total sales.
- Regional leader: West generated \$36.44M; Midwest was the lowest of the five regions at \$16.67M.
- City leader: New York was the highest-selling city at \$5.68M.
- Sales method: Online was the largest channel at \$44.97M, ahead of Outlet and In-store.
- Gender mix: Men contributed \$64.78M; Women contributed \$55.39M.

Important interpretation note

The Power BI dashboard contains a 79.9% Sales Growth KPI defined by the dashboard's selected comparison logic. The raw dataset's full-year 2021 sales total is approximately \$95.93M versus \$24.24M in 2020; this is a different comparison basis and should not be treated as the same growth measure.

Power BI Dashboard — Navigation & Executive Entry Point

9. Dashboard in Power BI

The Power BI solution is organized as a management-facing dashboard with a home page that provides navigation to four analytical pages. The design uses consistent KPI cards, page-specific color themes, filters, interactive visuals and drill-style navigation to move from high-level performance to detailed analysis.



Figure 1. Adidas Sales Analytics dashboard home page.

Page	Purpose
01 — Executive Overview	High-level KPIs, monthly sales trend, category mix, city performance and sales-method comparison.
02 — Retailer & Product Performance	Retailer contribution, category performance, regional comparison and sales contribution.
03 — Pricing & Profitability	Price vs units, price vs profit, category margins, profit per unit and sales-margin quadrants.
04 — Forecast & Scenario Analysis	Forecasted sales/profit plus interactive price and volume what-if scenarios.

Dashboard design objective

The home page is intended to answer “How is the business performing, where should I look next, and what can I explore?” before the user enters a specific analytical page.

Power BI — Executive Overview

10. Executive Overview

This page provides the management-level summary. It combines KPI cards with time-series, category, city and sales-method visuals so a decision-maker can understand the business without opening the detailed pages.

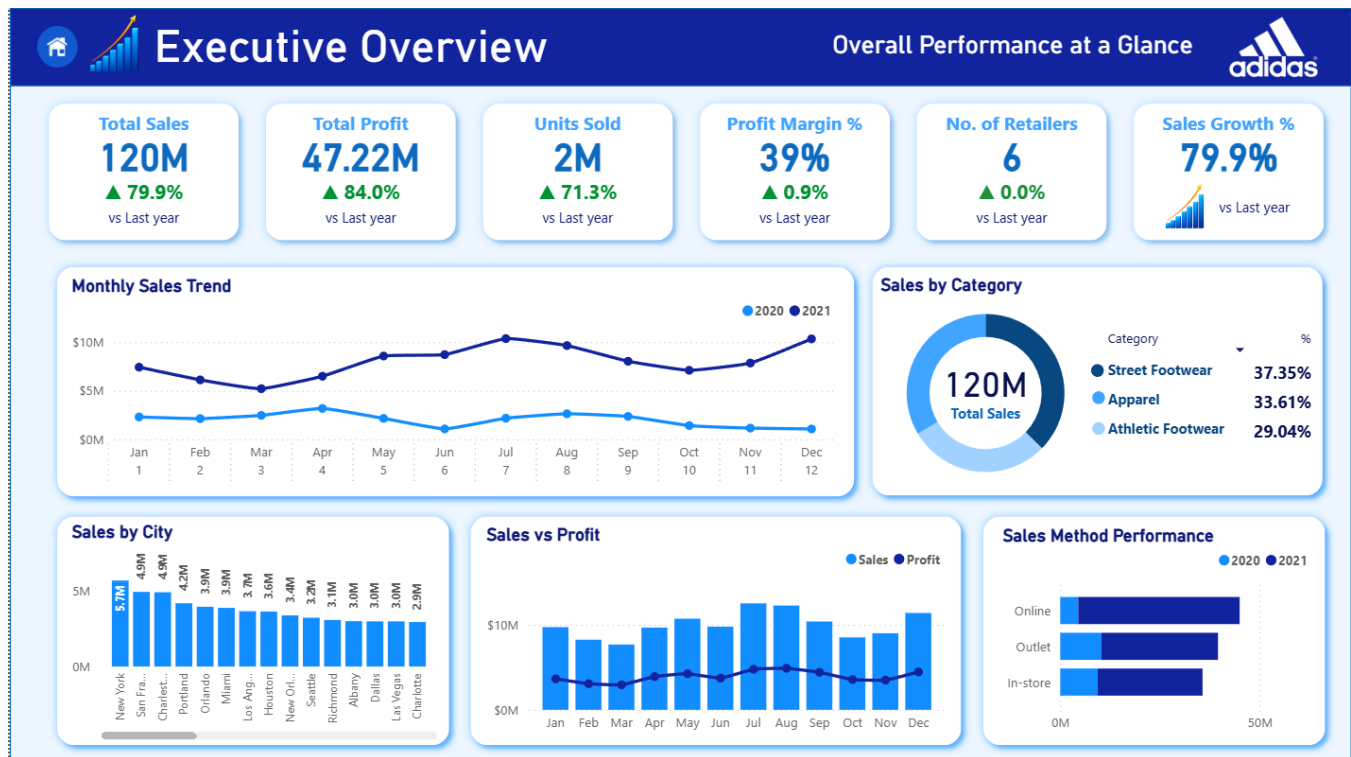


Figure 2. Executive Overview page.

- KPI snapshot: approximately \$120.17M sales, \$47.22M profit, 2.48M units and 39.3% weighted profit margin.
- Time trend: the dashboard compares 2020 and 2021 monthly sales and shows substantially stronger 2021 performance.
- Category mix: Street Footwear is the largest category at \$44.88M.
- Geography: New York leads city sales at \$5.68M.
- Channel: Online is the largest sales method at \$44.97M.

Management interpretation

The executive page is designed as the “first 60 seconds” view: confirm overall performance, identify the dominant category/channel, then move into the retailer, pricing or forecast page depending on the question.

Power BI — Retailer, Product & Pricing Analysis

11. Retailer & Product Performance



Figure 3. Retailer & Product Performance page.

- West Gear leads retailer sales at \$32.41M; Foot Locker and Sports Direct follow.
- The top three retailers contribute 71.6% of total sales, indicating meaningful concentration.
- West is the strongest region at \$36.44M, while Midwest is the lowest at \$16.67M.

12. Pricing & Profitability



Figure 4. Pricing & Profitability page.

- Aggregate margins are 40.4% for Street Footwear, 39.8% for Apparel and 37.4% for Athletic Footwear.
- The scatter plots evaluate relationships among price, units and profit, while the quadrant view separates high/low sales from high/low margin products.

Power BI — Forecast & Scenario Analysis

13. Forecast & Scenario Analysis

The final page extends the project from descriptive analytics into forward-looking decision support. It combines historical sales with forecast visualization and interactive what-if parameters for price and sales volume.

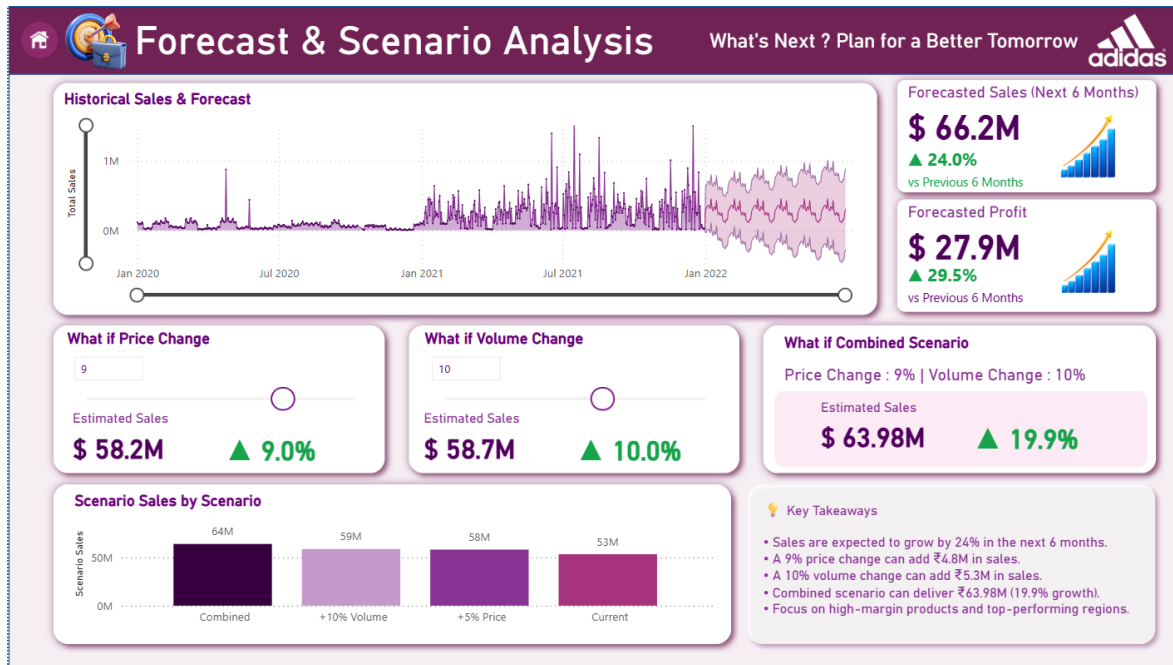


Figure 5. Forecast & Scenario Analysis page.

- Forecast: approximately \$66.2M sales and \$27.9M profit are shown for the next six months under the dashboard's current forecast configuration.
- What-If analysis: users can change price and volume assumptions and immediately see the estimated sales effect.
- Combined example: a 9% price change plus 10% volume change produces approximately \$63.98M estimated sales in the selected scenario.
- Decision value: scenarios allow management to compare assumptions before setting commercial targets.

14. Business Recommendations

- Protect and deepen relationships with the top three retailers because they contribute more than 70% of total sales.
- Prioritize high-margin, high-sales products and use online sales strength for channel optimization.
- Investigate the Midwest region for distribution, product mix and demand-generation opportunities.
- Use the What-If model before pricing or sales-target decisions, and extend the project with longer history, product forecasting and automated refresh.

Project conclusion

Excel → Python → PostgreSQL → Power BI → Forecasting → Business Action. The project demonstrates an end-to-end analytics workflow that converts transaction data into decision-ready insights.